



- 10 (a) Nerve impulses can be transmitted along a myelinated motor neurone to a neuromuscular junction at fast speeds of up to 100ms^{-1} .

Outline how a transmission speed of 100ms^{-1} is achieved by the neurone.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (b) When a nerve impulse reaches the neuromuscular junction, acetylcholine is released and diffuses to the sarcolemma.

Describe how the release of acetylcholine can result in the binding of calcium ions to troponin in the sarcomere.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

